Research Note

Influences of Hypoxia on Hatching Performance in Chickens with Different Genetic Adaptation to High Altitude

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ABSTRACT The experiments were conducted to assess how hatching performance is affected by chicken breeds and environment of high altitude and to analyze the vital factor of the low hatchability at a 2,900-m altitude. Eggs of Tibetan and Dwarf chickens were incubated at conditions of normobaric normoxia, normobaric hypoxia, hypobaric hypoxia, and supplemental O_2 at high altitude (hypobaric normoxia) during the whole incubation or at 0 to 7, 8 to 14, and 15 to 22 d of incubation, respectively. The results showed that the Tibetan chickens had greater hatchability (79.72%), lower water loss (12.90%), greater relative embryo weight (38.08%), and relative chick weight (68.41%) compared with the Dwarf chickens (31.69, 15.79, 30.71, and 65.21%, respectively) when both of them were incubated at a 2,900-m altitude. The hatchability was 71.60% in Tibetan chicken and 36.23% in Dwarf chicken under the normobaric hypoxia condition. The hatchability of chicken was efficiently increased with supplemental O_2. The previous results indicated that the O_2 deficit is the main factor resulting in the low hatchability and the poor chick quality of the lowland chicken breed when incubated at a 2,900-m altitude. Breeding chickens for adaptability to hypoxia and supplemental O_2 is a good way to improve the hatchability and chick quality at that altitude.

Key words: hypoxia, 2,900-meter altitude, genetic adaptation, hatchability

2008 Poultry Science 87:2112–2116
doi:10.3382/ps.2008-00122

INTRODUCTION

It has taken a long time to recognize that the reduction in the barometric pressure at high altitudes, with the attendant reduction in the O_2 partial pressure, makes the eggs incubated at high altitude subject to hypoxia, hypocapnia, and excessive water loss and results in retarded development and poor hatchability (Girard and Visschedijk, 1987). Previous studies conducted by simulating the conditions of 9,100-m and 5,500-m altitudes indicated that O_2 pressure, humidity, and concentration of CO_2 were the main factors that affect hatchability at high altitudes (Weiss, 1978; Visschedijk, 1985). However, there must be dose responses in the effects of hypoxia on embryonic development (Chen and Burggren, 2005). The vital factors should vary at different altitudes and in different species of birds. The altitude of 2,900 m represents a significant hypoxic challenge to the embryo (Chen and Burggren, 2005). Hatchability of chicken eggs would decrease dramatically at greater levels (Visschedijk, 1980). In Tibet, most areas are at about 2,900 m of altitude and the poultry industry is undeveloped. The Tibetan chicken (*Gallus gallus*) is an excellent indigenous poultry breed that inhabits this altitude. It is necessary to know the vital factor for chicken embryos to survive at a 2,900-m altitude. The purpose of this study was to examine the hatching performance of Tibetan chickens and lowland chickens incubated at conditions of normobaric hypoxia, hypobaric hypoxia, and hypobaric SO and analyze the vital factor of low hatchability at the altitude of 2,900 m.

MATERIALS AND METHODS

Experimental Design

Eggs were randomly collected from a flock of Tibetan chickens (**T**), which migrated to Beijing (100-m altitude) from Linzhi, Tibet (2,900-m altitude) for several years, and a single flock of Dwarf chicken (**D**). These breeding chickens were reared with same management and feeding regimens at the Experimental Farm of China Agricultural University in Beijing. A portion of eggs remained in Beijing, and the others were transported to Linzhi by airplane at room temperature for incubation test as design. Incubators used in this ex-

©2008 Poultry Science Association Inc.
Received March 21, 2008.
Accepted June 13, 2008.
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periment were the same commercial type (HJ-TL, Beijing Haijiang Incubator Ltd., Beijing, China). An automatic hypoxia simulation incubator (**AHSI**) designed by our laboratory was used in incubation under normobaric hypoxia conditions at lowland (Beijing) or supplemented O_2 (**SO**) treatment at highland (Linzhi). The AHSI was adapted from a commercial incubator (HJ-TL), was air-tight, and allowed automatic N_2 (99.99% N_2 high-pressure) or O_2 (100% O_2 high-pressure) injections to be maintained at either 15% O_2 (hypoxia) at lowland or 27.5% O_2 at highland. The 15% O_2 at sea level was approximately equivalent to the partial pressure of O_2 at a 2,900-m altitude, and the 27.5% O_2 at a 2,900-m altitude was approximately equivalent to normal partial pressure of O_2 at lowland. Carbon dioxide concentration in the air-tight AHSI was monitored and kept below 0.5% by automatic blowing into a container filled with potassium hydroxide that would adsorb CO_2. The O_2 and CO_2 concentrations were constantly monitored using an O_2 sensor (Alphasense Ltd., Essex, UK) and an infrared sensor (eSense-ip50, SenseAir, Delsbo, Sweden), respectively. Digital display and automatic command to gas injection were realized using 2 appearances (XSE/C-H, RST Sensor Ltd., Beijing, China). Relative humidity regulation in the AHSI was done by use of an external humidifier (E435DL, Yadu, Beijing, China).

The incubator rotated eggs every 2 h. All incubators were maintained at 37.8°C and a relative humidity of 60% at lowland or 65% at highland, which could decrease the increase of egg water loss induced by the decrease of barometric pressure during incubation. The eggs from T and C were exposed to one of the following protocols:

(1) Normobaric normoxia (**NN**): 20.9% O_2 for the duration of incubation at Beijing.
(2) Normobaric hypoxia (**NH**): 15% O_2 for the duration of incubation at Beijing.
(3) Hypobaric hypoxia (**HH**): incubation at Linzhi (2,900-m altitude), where the partial pressure of O_2 was equivalent to 15% of lowland air.
(4) Hypobaric normoxia or continuous SO (**CSO**): supplemented with 27.5% O_2 for the duration of incubation at highland.
(5) Early SO (**ESO**): SO from d 1 to d 7, then no SO for duration of incubation at highland.
(6) Middle SO (**MSO**): no SO from d 1 from d 7, then SO for d 8 to 14, then no SO for duration of incubation at highland.
(7) Late SO (**LSO**): no SO from d 1 to d 14, then SO for the duration of incubation at highland.

Measurement

All eggs were weighed, individually numbered, and randomly divided into different groups. At d 18 of incubation, all eggs were weighed again, candled, and those with evidence of a living embryo were transferred from the turning trays to hatching baskets. The eggs of d 18 were used to determine egg weight loss (**EWL**), defined as weight loss from d 1 to 18 to the egg weight (**EW**) × 100. At d 18, twelve eggs per group were used to determine the relative yolk-free embryonic body weight (**REW**), defined as the ratio of the yolk-free embryonic body weight to the EW × 100. When incubation was stopped, all hatched chicks were weighed to determine relative chick weight (**RCW**), defined as the ratio of chick weight to the EW × 100, and all unhatched eggs were opened to determine true fertility and the timing of embryonic mortality by visual appraisal as described by Lourens et al. (2006). Hatchability was expressed as the percentage of chicks to the true fertile eggs, and embryonic mortalities were expressed as the percentage of dead embryos to the true fertile eggs at the early (d 1 to 7), middle (d 8 to 14), and late (d 15 to 22) incubation periods.

Statistical Analyses

Data of embryonic mortality and hatching percentage were analyzed using a logistic regression model. The goodness of fit was estimated with the Homer-Lemeshow goodness-of-fit test. A GLM procedure of the statistical software package SAS version 8.2 (SAS Institute Inc., Cary, NC) was used to analyze EW, EWL, REW, and RCW in relation to breed and incubation conditions. When the means of the general model were statistically different, these means were further compared using Tukey's test. Significance was based on $P < 0.05$. The model was:

$$Y_{ijk} = \mu + \alpha_i + \tau_j + (\alpha\tau)_{ij} + \varepsilon_{ijk}$$

where Y_{ijk} = EW, EWL, REW, and RCW of egg k from breed i and incubation condition j; μ = the overall mean; α_i = the main effect of breed i (n = 2); τ_j = the main effect of incubation condition j (n = 7); $(\alpha\tau)_{ij}$ = the interaction between breed and incubator condition; and ε_{ijk} = the random error term. The values were expressed as mean ± SEM.

RESULTS

The results of the incubation experiment are shown in Table 1. Overall hatchability was greater in the T breed than in the D breed ($P < 0.01$). No significant difference was observed between the T-NN group and the D-NN group in hatchability or in mortalities of the early, middle, and late incubation period. However, because of greater mortalities of the early and late incubation period, the D-HH group had a lower hatchability compared with the T-HH group. The hatchability in the NH incubation significantly decreased from 86.49 to 71.30% in the T breed and from 84.94 to 36.23% in the D breed compared with the NN incubation. The NH incubation had similar mortalities and hatchability to the HH incubation within 1 breed of T or D. The hatch-

Table 1. Hatching performance in an incubation experiment[1]

Item	n	EW, g	Fertility, %	Mortality, % 0 to 7 d	Mortality, % 8 to 14 d	Mortality, % 15 to 22 d	Hatchability, %	EWL, %	REW, % (n = 12)	RCW, %
Tibetan chickens										
NN	288	47.37 ± 0.64^{b}	89.9^{b}	6.18^{b}	0.77^{b}	6.56^{d}	86.49^{a}	12.55 ± 0.21^{a}	45.11 ± 0.47^{a}	70.22 ± 0.31^{a}
HH	468	47.58 ± 0.56^{b}	91.7^{ab}	8.39^{b}	2.33^{ab}	9.56^{cd}	79.72^{b}	12.90 ± 0.17^{a}	38.08 ± 0.39^{f}	68.41 ± 0.34^{b}
NH	255	46.97 ± 0.69^{b}	90.2^{b}	10.00^{b}	0.44^{b}	18.26^{c}	71.30^{c}	12.38 ± 0.26^{a}	40.72 ± 0.43^{d}	69.69 ± 0.39^{ab}
CSO or HN	262	47.34 ± 0.82^{b}	92.8^{ab}	4.94^{b}	0.82^{b}	8.64^{cd}	85.60^{a}	12.87 ± 0.35^{a}	43.65 ± 0.44^{b}	68.96 ± 0.34^{b}
ESO	262	47.13 ± 0.87^{b}	93.5^{ab}	6.53^{b}	0.0^{b}	12.24^{c}	81.22^{ab}	12.56 ± 0.29^{a}	38.37 ± 0.36^{f}	68.58 ± 0.28^{b}
MSO	264	47.06 ± 0.78^{b}	90.9^{b}	7.92^{b}	1.25^{b}	12.50^{c}	78.33^{bc}	13.02 ± 0.31^{a}	41.90 ± 0.38^{c}	69.07 ± 0.35^{b}
LSO	264	47.42 ± 0.71^{b}	91.7^{ab}	8.26^{b}	0.83^{b}	8.26^{cd}	82.64^{ab}	12.48 ± 0.19^{a}	39.60 ± 0.41^{e}	68.62 ± 0.34^{b}
Dwarf chickens										
NN	256	56.79 ± 0.51^{a}	93.4^{aa}	6.28^{b}	0.84^{b}	7.95^{cd}	84.94^{ab}	13.67 ± 0.28^{b}	44.49 ± 0.39^{ab}	70.76 ± 0.27^{a}
HH	564	57.02 ± 0.53^{a}	94.0^{a}	17.17^{a}	3.58^{ab}	47.55^{a}	31.69^{f}	15.79 ± 0.34^{c}	30.71 ± 0.43^{i}	65.21 ± 0.49^{c}
NH	225	56.84 ± 0.77^{a}	92.0^{ab}	13.53^{ab}	1.93^{ab}	48.31^{a}	36.23^{f}	14.02 ± 0.22^{b}	31.83 ± 0.40^{h}	68.48 ± 0.47^{b}
CSO or HN	142	56.63 ± 0.82^{a}	91.6^{ab}	6.92^{b}	0.77^{b}	6.15^{d}	86.15^{a}	15.35 ± 0.24^{c}	41.57 ± 0.38^{cd}	65.36 ± 0.29^{c}
ESO	144	57.13 ± 0.91^{a}	95.4^{a}	8.09^{b}	2.21^{ab}	37.50^{b}	52.21^{d}	15.84 ± 0.32^{c}	31.70 ± 0.44^{h}	65.11 ± 0.36^{c}
MSO	145	56.81 ± 0.79^{a}	94.5^{a}	16.06^{a}	3.65^{ab}	39.42^{b}	40.88^{ef}	16.04 ± 0.35^{c}	37.73 ± 0.47^{f}	65.26 ± 0.35^{c}
LSO	145	56.72 ± 0.79^{a}	93.8^{a}	15.44^{a}	5.88^{a}	32.35^{b}	46.32^{de}	15.93 ± 0.33^{c}	35.50 ± 0.51^{g}	65.12 ± 0.36^{c}
Source of variation										
Breed		**	*	**	*	**	**	**	**	**
Treatment		NS	NS	**	*	**	**	*	**	**
Breed × treatment		NS	NS	**	NS	**	**	**	**	**

$^{a–i}$Means within a column lacking a common superscript differ ($P < 0.05$).

[1]EW = egg weight; EWL = egg water loss; REW = relative embryo weight at d 18 of incubation; RCW = relative chick weight at hatch; NN = normobaric normoxia incubation; HH = hypobaric hypoxia incubation; NH = normobaric hypoxia incubation; CSO or HN = continuous supplemented oxygen for the duration of incubation at highland (hypobaric normoxia incubation); ESO = incubation receiving supplemented oxygen at early period (d 1 to 7); MSO = incubation receiving supplemented oxygen at middle period (d 8 to 14); LSO = incubation receiving supplemented oxygen at late period (d 15 to 21).

*$P < 0.05$; **$P < 0.01$; NS = not significant.

ability in the CSO incubation increased up to 85.60% in the T breed and 86.15% in the D breed, which was similar to the value of NN incubation. No significant difference was observed in mortality and hatchability among the incubations of ESO, MSO, and LSO within the T breed. However, in the D breed, the ESO incubation improved the hatchability because of the decrease of mortality in early and late incubation periods, and the LSO incubation improved hatchability because of the decrease of mortality in the late incubation period, compared with the D-HH group. The D-MSO group also had greater hatchability than the D-HH group, but the difference was not statistically significant.

Overall, EWL was lower in the T breed than in the D breed ($P < 0.01$). Hypobaric condition (including the HH, CSO, ESO, MSO, and LSO groups) did not change EWL of the T breed significantly ($P > 0.05$) but increased the EWL of the D breed significantly ($P < 0.05$), compared with the normobaric condition (including the NN and NH groups).

Overall, REW was greater in the T breed than in the D breed ($P < 0.01$). The REW was 7.03% lower in the T-HH group (38.08%) than that in the T-NN group (45.11%) and was 13.78% lower in D-HH group (30.71%) than that in the D-NN group (45.11%). The NH incubation depressed the REW by 4.39% in the T breed and by 12.56% in the D breed compared with the NN incubation. The CSO incubation improved the REW by 5.57% in the T breed and by 10.86% in the D breed compared with the HH incubation. The increase of REW was bigger in MSO incubation than in ESO or LSO incubation and was bigger in LSO incubation than in ESO incubation.

The T breed had significantly greater RCW than the D breed at hypobaric conditions (including incubations of HH, CSO, ESO, MSO, and LSO), but they had comparable levels in normobaric conditions (including incubations of NN and NH). Supplemental O_2 incubations (including incubations of CSO, ESO, MSO, and LSO) did not change the RCW significantly compared with the incubation of HH in both T and D breeds, whereas there was improved hatchability in the D breed.

DISCUSSION

Embryonic development is a dynamic process determined by not only the genetic background of the organism but also by the environment in which it develops (De Smit et al., 2006). Visschedijk (1980) reported that hypoxia, excessive water loss, and hypocapnia are important factors for embryonic development and mortality at high altitudes. Chicken eggs laid at sea level and incubated at a high altitude have an increased incubation time and a lower hatchability (Visschedijk, 1980). However, it is still unclear which factor is the foremost to perturb embryonic development and how to improve the hatchability at a 2,900-m altitude.

The genetic potential of adaptation to high altitude was the important factor. The T breed was confirmed to have greater venous CO_2, hemoglobin concentra-

tion, and blood-O_2 affinity and a lower venous blood pH value, which worked for physiological adaptability to hypoxia (Gou et al., 2007; Wei et al., 2007; Zhang et al., 2007). Although the T breed used in this experiment had been migrated to lowland for several years, their hatchability remained at a high level (79.72%), and their REW were greater than that of the D breed when eggs were incubated at a 2,900-m altitude, which confirmed the results of the previous studies of Wei et al. (2007). Therefore, breeding chickens for the adaptability to hypoxia was a good way to improve the hatchability at 2,900-m altitude.

Gas exchanges are of fundamental importance for embryonic development during incubation and may affect the livability of the embryo (Onagbesan et al., 2007). The reduction in the barometric pressure and the O_2 partial pressure at high altitudes influences the gas exchanges of egg, and the combined effects consist of a lack of O_2 (hypoxia), CO_2 (hypocapnia), and water (dehydration; Visschedijk, 1991). In the present study, the incubation results from comparison between the D-NN group and the D-NH group and between the D-HH group and D-CSO group clearly indicated that O_2 deficit is the main factor causing the low hatchability and the low REW at a 2,900-m altitude, and supplementing O_2 up to 27.5% could significantly improve the hatchability and the REW at this altitude. Chronic hypoxia (15% O_2) caused an increase in mortality and a reduction in embryo body weight (Julian, 2000; Villamor et al., 2004). Therefore, hypoxia was a key factor that influenced the hatchability of lowland chicken eggs incubated at a 2,900-m altitude.

Previous studies suggested that chronic hypoxia during specific developmental critical phases in early or late incubation affected chick embryo survival, body weight, and caused developmental abnormalities (Altimiras and Phu, 2000; Dzialoswski et al., 2002; Chan and Burggren, 2005). Several studies suggested that a threshold of O_2 availability is required to initiate and sustain early embryo development (Ruijtenbeek et al., 2003; Sharma et al., 2006). Onagbesan et al. (2007) suggest that hypoxia at any level below 15%, no matter when applied but more especially during early development, alters the survivability of the chick, thus lowering hatchability. In the present study, we demonstrated that the D breed had high hatchability, receiving SO during early or late incubation, and had an increase of embryo weight, receiving SO during middle incubation, which was consistent with the report of Bagley and Christensen (1991). Therefore, it was suggested that SO during the early and late incubation period was important for improvement in hatchability at a 2,900-m altitude.

The low water loss in the T breed indicated that the eggs could avoid excessive water loss and CO_2 diffusion, which may be the part of the reason that the T breed had a greater hatchability and a greater REW than the D breed. Chicken embryo requires CO_2 at specific windows of development to enhance growth, early hatch, and optimum hatchability (Onagbesan et al., 2007). Several studies have shown that under high CO_2 concentrations during the first 10 d of incubations, embryonic growth was modestly accelerated, which was reflected in greater embryonic weight and a significant earlier hatching (De Smit et al., 2006, 2008; Bruggeman et al., 2007).

However, the data of the D breed incubation implicated that the increase of water loss or CO_2 diffusion was not vital to embryo survival. If the eggs received adequate O_2 during incubation at a high altitude, the D breed had a high hatchability; regardless, they suffered the increase of water loss (to 15.35%) and CO_2 diffusion (hypocapnia) because of the low pressure at a 2,900-m altitude. Handrich and Girard (1985) reported that compensation for water loss alone at a 2,900-m altitude does not counteract the reduction in hatchability. Girard and Visschedijk (1987) reported that altitude hypocapnia alone, without hypoxia and excessive water loss, had almost no significant effect on the development and hatchability of the embryo at a 2,800-m altitude. In the chicken embryonic development at that altitude, hypoxia, compared with hypocapnia, is the predominant respiratory factor. The partial pressure of CO_2 limits compatible with a normal embryonic development are wide, as are the limits for the ambient humidity and for the eggshell conductance (Carey, 1986).

Chick weight at hatch mainly depends on the initial EW and also on the amount of water lost during incubation (De Smit et al., 2006). In the present study, the RCW at hypobaric condition incubation were lower compared with chicks hatched at normobaric condition incubation, because the hypobaric condition increases the water loss from the eggshell, which was consistent with the report of Hassanzadeh et al. (2004). Some studies have shown that chronic hypoxia decreases the REW or RCW (Burton and Palmer, 1992; Altimiras and Phu, 2000; Dzialoswski et al., 2002; Rouwet et al., 2002; Sharma et al., 2006). But, on the other hand, Villamor et al. (2004) reported that hypoxia did not effect REW, and Blacker et al. (2004) found that hypoxia accelerated the developmental process of chick embryo. In the present study, hypoxic condition (NH) incubation decreased the REW and TCW in the D breed and only significantly decreased the REW in the T breed, compared with the NN incubation. The SO incubation did increase the RCW but significantly increased the REW in both T and D breeds, which indicated that the SO incubation at a 2,900-m altitude could improve the chick quality, then hatchability.

In conclusion, this study has shown that the T breed had a lower EWL and greater REW, RCW, and hatchability when it was incubated at a hypoxic condition. The O_2 deficit is the main factor influencing the hatchability and the chick quality of the lowland chicken breed when incubated at a 2,900-m altitude. The SO during the early and late incubation period was important for improvement in hatchability at this altitude. It

was indicated that breeding chickens for the adaptability to hypoxia and SO was a good way to improve the hatchability and chick quality at a 2,900-m altitude.

ACKNOWLEDGMENTS

The research was supported by 973 Project (project code 2006CB102101) of China and the Project of National Fundamental Platform for Scientific Work (project code 2005DKA21101).

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